

SECTION 15840

METAL HVAC DUCTWORK AND ACCESSORIES

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. This Section specifies the work necessary to furnish and install, complete, metal HVAC ductwork systems with pressure classifications ranging from negative 6 inches WC to positive 6 inches WC.
- B. The SMACNA publication HVAC Duct Construction Standard - Metal and Flexible generally governs the material gauges and the fabrication and installation techniques for systems specified in this Section. SMACNA criteria are superseded by the requirements of this Section.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for the referenced metal HVAC ductwork and accessories.
 - 1. Section 013300 – Submittal Procedures.
 - 3. Section 15200 – Vibration and Noise Control.
 - 4. Section 15840.01_dat – HVAC Ductwork Schedule.
 - 5. Section 15849 – Duct Hangers and Supports.
 - 6. Section 15990 – Testing, Adjusting, and Balancing.
 - 7. Section 230500 – Common Work Results for HVAC.
 - 8. Section 230700 – Equipment, Vessels, Piping, and Duct Insulation.

1.3 DESIGN STANDARDS

- A. SMACNA Publications
 - 1. HVAC Duct Construction Standards – Metal and Flexible.
 - 2. Fire Damper and Heat Stop Guide for Air-Handling Systems.
- B. NFPA Publications:
 - 1. 90A – Standard for the Installation of Air-Conditioning and Ventilating Systems.

1.4 SYSTEM DESCRIPTIONS

A. Definitions:

1. Duct: a tube or channel, including fittings, through which a gas moves.
2. Ductwork: a system of interconnected ducts and duct accessories, including, but not limited to, isolation and balancing dampers, smoke and/or fire dampers, access doors, turning vanes, flexible connections, hangers and supports, and all other miscellaneous items required for a completely operable and adjustable system.

B. Description:

1. Cleanroom and Spaces:

- a. Supply Ductwork (SA): ductwork and duct accessories from the discharge of the cleanroom makeup air handler to the connection of recirculation air ductwork or air handlers, final filters, diffusers, or free discharge.
- b. RA Return Air Ductwork (RA): ductwork and accessories from the termination or inlet to the inlet of the air handling unit.

2. Non-clean Areas:

- a. Supply Air Ductwork (SA): ductwork and duct accessories from the discharge of fans and air-handling units to their respective terminations, outlets, and free discharge openings.
- b. Return Air Ductwork (RA): ductwork and duct accessories from duct terminations, grilles, and air inlets to the inlets of the respective fans, return fans, and air-handling units.
- c. Exhaust Air Ductwork (EA): ductwork and duct accessories from duct terminations, grilles, and air inlets to the inlet of respective exhaust fans and from the discharge of the respective exhaust fans to the discharge termination to outdoors.
- d. Transfer Air Ductwork (TA): ductwork and duct accessories used to transfer air from a room of higher pressure to a room of lower pressure beginning at the grille or inlet of the air source and terminating at the grille, diffuser, or discharge outlet. Ductwork connected to a transfer fan is considered to be supply and return air ductwork as described above.

1.5 DESIGN CRITERIA

A. Supports, Anchorage, and Restraints:

1. General:

- a. When supports, anchorages, and restraints for equipment and supports and seismic restraints for ductwork are not shown on the Drawings, the Contractor shall be responsible for their design.
- b. Provide duct supports which are independent of other utilities.
- c. Connections to structural framing shall not introduce twisting, torsion, or lateral bending in the framing members. Provide supplementary steel as required.

2. Use the following table or the load limits published by the hanger components

manufacturer whichever is more stringent. The maximum total load per rod includes the trapeze, attachments, and insulation.

<u>Max. Total Load Per Rod (lb.)</u>	<u>Min. Hanger Rod (in.)</u>
610	3/8
1130	1/2
1810	5/8
2710	3/4
3770	7/8
4960	1
8000	1-1/4
11,630	1-1/2
15,700	1-3/4
20,700	2

1.6 QUALIFICATIONS

- A. Sheet metal work shall be fabricated and installed by qualified, experienced mechanics as specified herein and in accordance with the requirements of ASHRAE and the latest edition of the applicable SMACNA manual.

1.7 COORDINATION

- A. The Drawings do not attempt to show exact details of all ductwork. No extra payment will be allowed for obstruction by work of other trades or local obstructions to the work which require offsets. Where diagrams have been made to show duct connections, the Contractor is cautioned that these diagrams must not be used for obtaining material quantities.
- B. Changes in location of equipment or ductwork, advisable in the opinion of the Contractor, shall be submitted to the Engineer for review before proceeding with the work. All measurements and dimensions shall be verified at the site.
- C. Duct sizes shown on the Drawings represent the nominal free area required for that service. Where changes in duct dimensions are necessary to coordinate the installation, the Contractor may change to an equivalent duct area with different dimensions as determined using a Trane Ductulator or Carrier or ASHRAE equivalence charts, after approval of the Engineer. Aspect ratio shall not exceed 3 to 1 unless approved by engineer of record.
- D. Coordination with Existing Conditions and with Other Trades:
 - 1. Coordinate the installation of ductwork with existing conditions and the work of other trades to allow the installation of ductwork, as well as proper access and operation of dampers and operators.
 - 2. Where existing thread rod, strut material, miscellaneous supports, conduit, or piping under 1-inch diameter obstructs the passage of the ductwork, they shall be relocated by the Contractor at no additional cost to the Owner. Coordinate the work with other

trades.

1.8 SUBMITTALS

A. Provide the following with the Bid

1. Product data for duct accessories.

B. Provide the following within 3 weeks of award of Contract

1. Shop drawings of suspended ductwork indicating point load locations, along with applicable details keyed to layouts.
2. Shop drawings for support frames, equipment and ductwork supports, and associated anchorage indicating point load locations, along with Engineer's calculations and details keyed to the layouts.

C. Provide a certificate of test approval by Owner's representative on all systems within 1 week of test date.

1.9 QUALITY ASSURANCE

- A. Applicable Codes and Standards: The materials and methods of construction for sheet metal ductwork and accessories hereinafter specified shall comply with the current editions (unless specifically indicated otherwise) of the practices, methods or standards prepared by the applicable technical societies and associations listed in Section 15023 – Codes and Standards of these specifications.
- B. Unless indicated on the Drawings, or specified, construction shall be in accordance with 6" pressure class requirements of "HVAC DUCT CONSTRUCTION STANDARDS, METAL AND FLEXIBLE" published by Sheet Metal and Air-Conditioning Contractor's National Association, Inc. (SMACNA), 2005, hereinafter referred to in this Specification as Duct Manual. Should conflict occur between this manual and its latest edition, the more stringent requirement shall take precedence.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Ductwork Materials:

1. Galvanized Steel (GS): hot-dipped GS sheet, lock-forming grade, conforming to ASTM A525 and A527, having G90 zinc coating in conformance with ASTM A90, unless heavier galvanizing is specified.
2. Hot-Rolled Steel (HRS): HRS sheet, commercial quality, conforming to ASTM A568 and A569, or A366 HRS plate with ASTM A36 hot-rolled structural steel shapes.
3. Refer to Section 15840.01_dat for metal ductwork gage and pressure requirements by service.

B. Sealants and Tapes:

1. Duct Sealer - Acceptable Manufacturers:

- a. 3174 United Duct Sealer.
- b. Vulkem (one part polyurethane) #116 Sealant (exterior of building only).
- c.. Vulkem (one part polyurethane) #921 Sealant (exterior of building only).
- d. Pecora BC 158 Butyl Sealer.
- d. Seal Bond 95
- e. Sikaflex 1A, for use in cleanroom

2. Gaskets:

- a. United McGill – Butyl tape
- b. United McGill Uni-Prene Neoprene

3. Duct Tape:

Duct tape shall only be used down stream of VAV type systems.

Acceptable Manufacturers:

- a. United Duct Tape.
- b. NASHUA No. 357 silver cloth tape.

- 4. Shrink Band (Round Ducts Only) – Acceptable Manufacturer: Raychem.
- 5. Silicon Sealant (SS Ductwork Only) – Acceptable Manufacturers:

- a. Clear GE Construction 1200.
- b. Dow 999.

2.2 RECTANGULAR HVAC DUCTWORK

- A. Materials: Construct rectangular HVAC ductwork and all components and accessories from materials specified above as scheduled in Section 15840.01_dat – HVAC Ductwork Data Sheets except as noted on Drawings.

B. Fabrication:

- 1. General: Fabricate ductwork from sheet material equaling or exceeding minimum wall thickness and reinforcing as scheduled in the SMACNA rectangular duct construction schedules to comply with duct pressure classifications specified. Cross break or bead all duct widths over 12 inches and horizontal surfaces to prevent ballooning or breathing.
- 2. Fittings:
 - a. General: Fabricate fittings for easiest airflow.
 - b. Radius Elbows: Fabricate with centerline radius equal to or greater than one and a half times the duct dimension in the plane of the turn.

- c. Mitered Elbows: Provide with turning vanes as specified under ductwork accessories.
- d. Branch Takeoffs, Rectangular to Rectangular:
 - 1) 3 Inches WC and Greater Pressure Class: Use parallel flow branches where possible. Where branch taps must be made, use 45 degree entry with $L = 1/4, 4$ inches minimum.
 - 2) 2 Inches WC and Lower Pressure Class:
 - a) Use parallel flow branches where possible. Where branch taps must be made, use 45 degree entry with $L = 1/4w, 4$ inches minimum.
- e. Joints:
 - 1) Longitudinal: Pittsburgh lock flooded with mastic prior to assembly. Snaplock is unacceptable.
 - 2) Transverse: Use demountable joint system, such as Ductmate, Nexus, or TDC on rectangular ductwork except where it is impractical. Seal corners prior to assembly.

2.3 ROUND SINGLE-WALL DUCTWORK

- A. Materials: Construct round HVAC ductwork and all components and accessories from materials specified above as scheduled in Section 15840.01_dat – HVAC Ductwork Data Sheets.
- B. Round ducts shall be fabricated in accordance with appropriate tables, figures and requirements of the Duct Manual. Unless pressure classifications are specified or noted otherwise, duct systems shall be constructed in accordance with 2 inch W.G. pressure for recirculating system, 6 inch W.G. for make-up air unit requirements.
- C. Ducts shall be round and made of galvanized steel sheets, as applicable. Duct flanges and reinforcing angles shall be galvanized coated ASTM A36 hot rolled steel for galvanized ducts. Round ductwork shall be fabricated in 12 foot maximum sections.
- D. Longitudinal seams shall be either continuous welded, grooved and sealed with sealing compound after assembly, or machine formed spiral lockseam. Transverse joints shall be either welded flange, bolted companion angle with 1/8" neoprene gasket, or slip joints having beaded sleeve coupling sealed with either gray shrink bands, as manufactured by Raychem Corporation, or an approved sealer. Sheet metal screws shall not be used. Slip joints shall be installed in accordance with spiral duct manufacturer's written instructions.
- E. Elbows in round ductwork shall be a minimum of two gauges heavier than straight ducts of equal diameter. Centerline radius of elbows in round ductwork shall be 1-1/2 times the duct diameter if no centerline radius is shown on the Drawings, and shall be made with the

following order of preference:

1. Smooth, two section, stamped.
 2. Five (5) piece, with welded seams – Greater than 45°.
 3. Three (3) piece, with welded seams – 45° or less.
- F All fittings are to have continuous welds along all seams. All divided flow fittings are to be manufactured as separate fittings, not as tap collars welded into spiral duct sections.
- G All 90 degree tees and 45 degree laterals (wyes) up to and including 12" diameter tap size shall have a radius entrance into tap produced by machine or press forming. Entrance shall be free of weld build-up, burrs, or irregularities.
- H Branch run-outs to be conical type

2.4 DUCT ACCESSORIES

- A. General: Fabricate duct accessories from the same material as the ductwork in which the accessory will be installed, unless specified otherwise.
- B. Manual Splitter Dampers: fabricated from 18-gauge metal of the same material as the ductwork with splitters long enough to completely close the branch duct without flutter. Provide a locking quadrant and align its operating handle with the damper blade.
- C. Manual Opposed-Blade Balancing Dampers (shall match duct pressure class):
1. Acceptable manufacturers fabricate in accordance with SMACNA low pressure duct constructions standards, except as detailed on the Drawings and as specified.
 - a. Air Balance
 - b. Nailer-Hart industries
 - c. Prefco Products
 - d. Ruskin
 2. Blade, rod, linkage, frame, and fastener material shall be the same as the ductwork in which the damper is to be installed.
 3. Rod shall be of sufficient length and the quadrant base shall be offset a distance from the duct to allow the ductwork to be insulated, without interfering with the operation of the damper handle.
 4. Blades shall be center crimped for stiffness and to receive pins and rods and edge crimped to interlock with adjacent blades on full shutoff. Provide additional stiffening angles, as required.
 5. Provide a prime-coated or galvanized channel frame for mounting of bearings with angle stops on top and bottom.
 6. Provide continuous rods on all motorized damper assemblies. Mark the ends of the damper shaft extension with a saw-cut to indicate blade position.

7. Linkage and hardware shall be SMACNA-accepted as manufactured by Ventlok, arranged for gang operation and opposed-blade action without jamming or racking.
8. Bushings for GS dampers shall be oil-impregnated bronze, sized to match with pins. Bushings for aluminum and stainless steel ductwork shall be nylon.
9. Blades shall be a minimum of 18-gauge steel or equal strength aluminum. Maximum blade width shall be 12 inches. Provide multisection dampers for dampers over 48 inches wide.
10. Provide specified locking quadrants for ductwork serving nonclean areas. Provide specified dial regulators for ductwork serving cleanrooms.

D. Regulators and Quadrants:

1. Acceptable Manufacturers:
 - a. Ventlok as specified.
 - b. Duro-Dyne Specline/Quadline equivalent.
2. Concealed Accessible Uninsulated Ductwork:
 - a. Ventlok Figure 635 dial regulator and 607 end bearing.
 - b. Ventlok Figure 555 locking quadrant and 607 end bearing.
3. Concealed Accessible Insulated Ductwork:
 - a. Ventline Figures 637 or 639 dial regulator and 607 end bearing to match insulation thickness.
 - b. Ventline Figure 555 locking quadrant with an elevated standoff and 607 end bearing of the same height as the insulation thickness.
4. Concealed Inaccessible Ductwork: Ventlok Figure 677 concealed damper regulator with plain cover, extended rod, and universal joints and miter gears as required.
5. Exposed Uninsulated Ductwork in Finished Spaces or Cleanrooms: Hi-Vel Ventlok Figure 640 dial regulator with 609 end bearing.
6. Insulated Ductwork Serving Cleanrooms:
 - a. Ventlok Figure 641 dial regulator with Hi-Vel hardware and Figure 609 end bearings. Trim and tape insulation as required for full stroke of damper handle.

E. Turning Vanes:

1. Acceptable Manufacturers:
 - a. Elgen All-Tight.
 - b. Duro-Dyne Type yR.

2. Description: hat channel or embossed vane side rails with shop-fabricated, single-blade turning vanes of the same material as the ductwork.

F. Flexible Connectors:

1. Install flexible connections where shown on the drawings. Install with a minimum of 1-1/2 inch slack in flexible material. Connections shall be approximately 6 inches long, of which not less than 3 inches shall be flexible. Connections shall be substantially airtight and made to be removed and replaced easily.
2. Acceptable Manufacturers:
 - a. Ventfabrics Ventlon.
 - b. Duro-Dyne Durolon.
3. Description: neoprene-coated, fire-retardant woven glass fabric, minimum 36-ounce-per-square-yard density, UL listed, crimped into metal edging strip with 2-inch minimum clearance between equipment casing and ductwork, unless otherwise indicated.

G. Duct

1. Access Doors: Acceptable Manufacturers:
 - a. Ventlok.
 - b. Duro-Dyne.
 - c. Ruskin.
2. Description:
 - a. Double-wall access door with minimum 1-inch-thick insulation between walls, angle, or Z-frame, sealing gasket around the periphery, double layer Plexiglas vision panel with air space between, hinged with Ventlok No. 100 quick-release latches. Access doors smaller than 12 inches square may be secured with sash locks.
 - b. Sizes shall be as follows:

Duct Size	Frame Size	Number of Hinges	Access Requirements
6 through 8 inches	6 by 8	Two	Handhole
10 inches	10 by 12	Two	One hand and sight
12 through 16 inches	12 by 16	Two	Two hands and head sight
Greater than 16 inches	16 by 24	Three	Body entry

- c. Provide sufficient sealant for application between the door frame and duct for installations in ductwork with seal Classes A and B and in ductwork serving cleanrooms.
- d. Patch plates attached with screw fasteners are prohibited, except in locations

where they are specifically noted.

H. Backdraft Damper (BDD-1):

1. Acceptable manufacturers:
 - a. American Warming and Ventilating.
 - b. Dowco.
 - c. Ruskin.
2. Description:
 - a. Commercial-quality, multiblade, parallel-action, counterbalanced backdraft dampers of 16-gauge GS or extruded aluminum with center-pivoted blades of maximum 6-inch width with felt or flexible vinyl-sealed edges, linked together in rattle free manner with 90 degrees stop, steel ball bearings, and plated steel pivot pin. Channel frame.
 - b. Gravity backdraft dampers smaller than 18 by 18 inches furnished with air-moving equipment may be the air-moving equipment manufacturer's standard construction.
3. Ratings: 3,500-fpm maximum velocity, 1 inch WC backpressure, set to relieve above 0.02 inch WC differential pressure.

I. Sheet Metal Plenums: Fabricate from minimum 18-gauge metal of the same material as the ductwork, braced with steel framing for rigidity. Provide man-access doors for plenums size 6 feet or greater in height.

J. Pneumatic Actuators – Provided by Division 17.

K. Louver Blank-Off Sections: Fabricate from 20-gauge galvanized steel sheets lined with sound attenuation material as specified in Section 230700 – Equipment, Vessels, Piping, and Duct Insulation. Attachment to louver shall be airtight and installed in such a manner as to allow easy removal. Prime and paint the outside face of blank-off sections with two coats of flat black exterior paint.

L. Contractor-Fabricated Roof Hood:

1. General: Fabricate air inlet and exhaust hoods in accordance with SMACNA HVAC Duct Construction Standards - Metal and Flexible.
2. Materials: galvanized steel, minimum 18-gauge base and 20-gauge hood; or aluminum, minimum 16-gauge base and 18-gauge hood; suitably reinforced with removable hood, bird screen with 1/2-inch square mesh for exhaust and 3/4 inch for intake, and factory baked enamel finish.
3. Insulate inside surfaces of hoods not connected to duct systems.
4. Accessories: minimum 16-inch-high curb base with insulation between duct and

5. curb.
5. Performance: Size hood outlet area to exceed twice the throat area.

M. Contractor-Fabricated Gooseneck:

1. General: Fabricate goosenecks in accordance with SMACNA HVAC Duct Construction Standards - Metal and Flexible.
2. Materials: minimum 18-gauge galvanized steel.
3. Accessories: minimum 16-inch-high curb base where size exceeds 9 by 9 inches.

N. Access Doors:

1. Access doors are to be installed in ducts, casings and housing where shown on the Drawings and as required for inspections and maintenance. Unless otherwise noted, access doors are to be made of same material as ducts, casings, or housings in which they are installed. Access doors to be fabricated with insulation equal to insulation of ductwork or housings in which they are installed.
2. Access Doors for Duct:
 - a. Access doors shall be installed in a position in ductwork that insures ease of access, cleaning and serviceability.
 - b. Install access doors in ducts at each splitter damper, volume damper (manual & motorized), and fire damper; and as shown on the Drawings.
 - c. Access doors for rectangular ductwork shall be fabricated in accordance with Fig. 2-12 of the Duct Manual. Frames shall be Frame 2, as shown in Fig. 2-12. Latches shall be Lock Type 2, as shown in Fig. 2-12. They shall be #100 Ventlok Latch, by Vent Fabrics, Inc. Hinges shall be #370 Ventlok Hinge, by Vent Fabrics, Inc. Provide 1" x 1" x 16 gauge angle frame (same material as duct) riveted to exterior surface of duct around door opening.
 - d. Unless noted otherwise, access doors for rectangular ductwork shall be furnished as follows:

<u>Duct Dimension</u>	<u>Minimum Door Size</u>
Up to 18"	8" W x 12" L
19" thru 46"	16" W x 12" L
47" and Over	24" W x 18" L

- e. Access Doors for round ductwork shall be United McGill Corp. Type AR2-I combination access opening and vacuum relief having non-insulated galvanized steel frame and insulated galvanized steel door with chain retainer, or an approved equal.
3. Access Doors for Casings and Plenums:
 - a. Construct doors in accordance with Fig. 6-11, Detail E of the Duct Manual,

except use Ventlok No. 310 handles and Ventlok No. 154 T-Hinges.

- b. Doors and door frames are to be made of structural shapes and/or formed sheet metal and shall be strong and rigid enough to open and close properly and to maintain their air sealing properties under operating conditions. Doors are to be insulated if adjacent sheet metal is to be insulated. Insulated doors are to be hollow metal construction with insulation inside sealed metal shell. Uninsulated doors may be single thickness metal, properly reinforced for stiffness and strength. Wherever possible, doors are to open against air pressure so air pressure will tend to hold door closed. Provide structural steel angle at each side of door opening, extending from bottom to top of casing. Around each door opening in insulated surfaces, provide a 20 gauge flange, 3" wide, extending over insulation. Minimum size of access doors for casings and plenums shall be 16" wide by 24" high, or as specified.

O. Welding:

1. Preparation for Welding: Dirt, grease, forming or machining lubricants, or any organic material shall be removed from areas to be welded by cleaning with a suitable solvent or by vapor degreasing. Oxide coating shall be removed just prior to welding. This may be done by etching or scratch brushing. Suitable edge preparation to assure 100% penetration in butt welds shall be used.
2. Cutting: Material may be sheared, sawed, cut with a router, or arc cut. All edges which have been cut by arc process shall be planed to remove edge cracks. Cut edges shall be true and smooth, and free from excessive burrs or ragged breaks. Re-entrant cuts shall be avoided wherever possible. If used, they shall be filleted by drilling prior to cutting. Oxygen cutting of aluminum alloys shall not be permitted.
3. Welding Procedure: Duct shall be welded with an inert gas shielded arc or resistance welding process. No welding process that requires a welding flux shall be used. Other filler metal alloys may be used provided they meet qualification test requirements, and result in welds with listed alloy filler. Pre-heating for welding is permissible, provided temperature does not exceed 400°F for a total of 30 minutes.
4. Re-welding Defects: Portions of joints that have been rejected on inspection because of defects may be repaired by re-welding only. Defective area shall be removed by chipping or machining. Flame cutting shall not be used before re-welding. Joints shall be inspected to assure that all defective weld metal has been removed and that joint is accessible so that welding operator can obtain full penetration through joint.

P. Roof Curbs: Where ductwork penetrates building roof, roof curbs shall be provided by Contractor; Design and insulation to be reviewed and approved by MTI and trades.

Q. Duct Filter Housing and Filters:

1. Acceptable manufacturers:

- a. Flanders. Surepleat.
 - b. Camfil.
- 2. General: Provide duct filter housing designed to accommodate ASHRAE-rated filters 2in. or 4 in. depth as specified in design documents. Duct filter housing to be factory-assembled flanged units suitable for installation in a duct or next to an air handling unit. Housing to accommodate filters for different types and efficiencies up to 95% per ASHRAE Standard 52.1 dust-spot test methods ASHRAE 52.2 Test Standards. Filters to be easily accessed through access doors. Factory-gasketed channels and doors assure positive sealing. Filters will slide into frame. Position housing in accessible location.
- 3. Materials and Construction: Housing to be constructed of 16ga. galvanized steel with upstream and downstream outwardly turned flanges. Units include "Z" channel vertical support members on all four corners. Metal-to-metal joints shall be sealed with silicone compound. Filter racks to be aluminum with replaceable gasketing. Housing to be equipped with two quick opening access doors that are sealed on the perimeter. Doors to be equipped with door knobs. Filter rack to accommodate 24"x24", 12"x24" or 24"x12" nominal size filters or as specified in design documents. Filter housing to withstand system static pressure range of - 4.0 in. w.g. to +3.0 in. w.g. Independently support filter housing from connecting ducting.
- 4. Filters: Provide ASHRAE-rated filters with 2in. or 4in. depth as specified in design documents. Filters to comply with ASHRAE Standard 52.1 dust-spot test methods ASHRAE 52.2 Test Standards. Provide (2) sets of disposable pleated air filters.

2.5 DUCT HANGERS AND SUPPORTS

A. General:

- 1. Duct hanger and support material gauges shall comply with details on the Drawings. Where details are not provided, comply with SMACNA HVAC Duct Construction Standards - Metal and Flexible.
- 2. Suspension material may be threaded rod, metal strap, structural shapes, or strut material. The use of wire as suspension material is prohibited.

B. Upper Attachments:

- 1. Concrete Construction:
 - a. Cast-in-place inserts as specified in another division of work are acceptable.
 - b. Expanding concrete anchors as specified in another division of work may be used.
 - c. The use of powder-actuated fasteners is prohibited without prior written

consent by the Engineer.

2. Steel Construction:

a. Acceptable Manufacturing Company.

- 1) Fee and Mason Manufacturing Company
- 2) Grinnell.

b. Support ductwork from structure or supplementary steel, utilizing bean clamps designed for the intended service. Support from metal decks above is prohibited. Provide supplementary steel as detailed and as required.

C. Trapezes: Utilize structural steel shapes or strut material. Separate dissimilar duct and support materials with 1/16-inch rubber sheet.

D. Support Frames: structural steel shapes or strut frames for support of ductwork, accessories, and HVAC system components with nuts, bolts, fittings, etc., compatible with the frame material.

PART 3 - EXECUTION

3.1 FABRICATION AND INSTALLATION

A. General:

1. Duct sizes shown on the Drawings are inside duct dimensions and represent the net-free area of the duct.
2. Install the entire duct system with a minimum number of bends and transitions.
3. Except where Raychem are applied, seal all nonwelded joints with a specified chemical-resistant mastic sealant while joining.
4. Install additional reinforcement to preclude ballooning or breathing of ducts.

B. Purchased Components: Install ductwork components, such as Duct-Mate, etc., and accessories in accordance with the manufacturers' instructions.

C. Joints:

1. Assemble round and oval sections using couplings with duct sealer mastic applied to inside and outside of section. After applying mastic, push sections together with 1-inch minimum overlap. Apply a coat of duct sealer mastic over joint and coupling.
2. At the Contractor's option, sections may be assembled with all joints brazed or welded. If welded, internal area affected by weld must be cleaned and coated with galvanic paint.

3. Raychem shrinkwrap may be used in lieu of mastic and tape system with prior approval.
- D. Branch Takeoffs:
1. Takeoffs to terminal units shall be the same free area as terminal unit inlet or as shown, whichever is larger.
 2. Connect rigidly to terminal unit inlet without flexible duct.
- E. Splitter Dampers: Installation of splitter dampers is restricted to only those locations shown.
- F. Balancing Dampers:
1. Install where shown and where required. Install quadrants or regulators as specified, leaving all dampers locked wide open.
- G. Turning Vanes: Install in all 90 degree miter elbows.
- H. Flexible Connectors: Make connections to fans and other rotating equipment using flexible connectors with 2 inches minimum clearance between casing and ductwork. Flexible connectors are not required on internally spring-isolated units equipped with internal flexible connectors unless shown on Drawings.
- I. Ducts Through Walls and Floors:
1. Caulk nonfire-rated openings with Minnesota Mining and Manufacturing Company, 3M butyl sealer.
 2. Seal openings through fire-rated partitions with fire sealants as specified in Section 07900 - Joint Sealers.
 3. Attach sheet metal collar to duct to neatly cover exposed openings; collar shall tightly fit to surfaces.
- J. Connections to Equipment:
1. Install flanges to match those of connecting factory-fabricated equipment.
 2. Make transitions in ductwork from rectangular to round or to accommodate the sizes of factory-fabricated equipment, e.g., coils and dampers using tapered ducts, not by safig.
- K. Duct Access Doors: Provide duct access doors for inspection and maintenance at each fire damper, smoke or ionization detector, electric duct heater, coil, filter, humidifier, automatic damper, fan plenum, and elsewhere as indicated.
- L. Backdraft Dampers:

1. Provide backdraft dampers on exhaust fans or exhaust ducts nearest to outside and where indicated.
2. Provide duct access doors for checking operation and maintenance.

M. Fire Dampers & Combination Fire/Smoke Dampers:

1. Install dampers in accordance with NFPA 90A, UBC, the manufacturer's written recommendations, and the product's UL listing.
2. Install fire dampers at locations indicated, where ducts and outlets pass through fire-rated construction, and where required by authorities having jurisdiction.
3. Install with required perimeter mounting angles, sleeves, breakaway duct connections, corrosion-resistant springs, bearings, bushings, and hinges.
4. Install an access door in a serviceable position on the link or latch side of the damper of sufficient size to allow resetting of the damper and replacement of the link. Doors shall be compatible with the pressure rating of the duct in which they are installed.
5. Demonstrate resetting of smoke/fire dampers to authorities having jurisdiction and to the Engineer.

N. Painting:

1. Paint supports in accordance with Section 15048 - Mechanical Painting.

O. Vibration Isolators:

1. Vibration isolation shall be in accordance with Section 15200 - Vibration and Noise Control, and as detailed on the Drawings.

P. Leak Testing: See Section 15990 Testing, Adjusting, and Balancing for additional requirements.

Q. Make branch connections in accordance with the Duct Manual, and details indicated on the Drawings.

R. Ductwork of different materials to be connected by gasketed flanges.

S. For duct, fittings, and installation methods not otherwise specified, United Sheet Metal Products and Methods to be standard of quality to determine equivalency.

3.2 DUCT HANGERS AND SUPPORTS

A. General Support Locations:

1. Install hangers close to transverse joints of main ducts and branches, clinch collar

- branch connections, and the first branch elbows after nested splits.
2. Locate hangers of ducts penetrating walls or partitions as though the walls will contribute no support to the duct.
 3. Install hangers in pairs on exact opposite sides of duct. Install duct supports from floors and concrete walls when necessary.
 4. Maintain hanger spacing intervals not greater than the specified maximums.
 5. Install hangers at the midpoint of small- and medium-sized horizontal-vaned square elbows. On wide-vaned square elbows, install additional hangers at maximum allowable intervals (or less) measured along the heel lines of the elbows.
 6. Provide a set of hangers at the midpoint of small- and medium-sized horizontal radius elbows with greater than 20 degree offset. Install one or more supplementary hangers along the inside and outside arcs of large-radius elbows of any angle whenever the lengths of these arcs exceed the maximum hanger spacing length for that duct size.
 7. Provide at least one set of hangers for short duct branches 3 feet or less in length.
 8. Provide each duct riser with a minimum of two supports completely spanning the shaft opening at each floor. One pair of supports may be used to support more than one duct riser, provided that the strength of the supports is increased appropriately and proper additional supplementary steel is used at the extra risers.
 9. Support duct risers located between floors that are more than 15 feet high at each floor and halfway points between floors. The distance between intermediate supports on very high floors should not exceed 12 feet for round ducts or 10 feet for rectangular ducts (intermediate hangers may be supported from an adjacent concrete wall or hung by rods from supports on the floor above).
 10. Provide one or more sets of hangers for equipment in duct runs, such as heating coils, air terminal units, mixing boxes, etc., as recommended by their manufacturers.

B. Specific Support Locations - Locate Duct Hangers:

1. Approximately 2 to 24 inches from flexible connections of fans.
2. Approximately 2 to 24 inches from the outlets or flexible connections of terminal units or mixing boxes.
3. Approximately 12 to 36 inches from the main duct to the first hanger of long branch ducts.
4. Approximately 2 to 12 inches from the ends of all branch ducts and linear diffuser plenums.
5. Approximately 2 to 24 inches from fire damper breakaway joints.
6. Approximately 2 inches to half the duct width, plus 2 inches from the vertical centerline of the lower elbow of short vertical offsets made with vaned square elbows. The width refers to the dimension of the elbow in the plane of the turn. The heavier the duct, the closer the hanger should be to the centerline of the elbow.
7. Approximately 2 inches to half the duct width, plus 2 inches from the vertical centerline of the bottom and top elbows of vaned square elbow offsets over 8 feet high. Diameters of hanger rods of heavy offsets must be increased proportionately. The heavier the duct, the closer the hanger should be to the centerline of the elbow.

8. Approximately one-eighth of the arc in from the ends of bottom and top radius elbows of vertical offsets longer than 8 feet. Short vertical offsets require hangers at the bottom elbow. Likewise, sloping offsets need at least one set of hangers at their lower-radius elbow. Fasten hangers on sloping or arched surfaces to ducts to prevent slippage.
9. Approximately 6 to 12 inches from transverse joints of ducts whose lengths are the same as specified hanger intervals.
10. Approximately 6 to 12 inches from one side of walls or partitions penetrated by ducts.

C. Support Spacing - Maximum Permitted Hanger Spacing:

1. Provide duct support hanger spacing in accordance with 2005 SMACNA Manual or as noted in associated specification section. No spacing is to exceed 10 feet.

D. Support Method:

1. Support low ductwork from floors or other steel structures acceptable to the Engineer using struts or structural steel shapes or bars. Where possible, maintain 6 inches clear between the bottom of the duct and the supporting structure.
2. Support ductwork suspended from overhead using trapeze supports for rectangular ducts or straps for round ducts with the hanger strap wound completely around the duct. Use threaded rod for support from above.
3. Use of sheet metal screws for attaching support straps to ducts is prohibited.
4. Trim support material to minimum length after elevation is adjusted.

- E. Use the following table or the load limits published by the hanger components manufacturer whichever is more stringent. The maximum total load per rod includes the trapeze, attachments, and insulation.

<u>Max. Total Load Per Rod (lb.)</u>	<u>Min. Hanger Rod (in.)</u>
610	3/8
1130	1/2
1810	5/8
2710	3/4
3770	7/8
4960	1
8000	1-1/4
11,630	1-1/2
15,700	1-3/4
20,700	2

- F. Refer to Section 15849 – Duct Hanger and Supports for further specification on Hangers and Supports.

3.3 TEST AND ADJUST

A. Pressure Testing of HVAC Ductwork:

1. Description:

- a. Prior to connection to equipment and before applying insulation, perform a dynamic pressure test on HVAC ductwork systems using a high-pressure blower with a calibrated orifice and manometer. Provide necessary blowers, gauges, connections, and similar items required to perform the tests.
- b. Repair all leaks and retest until stipulated results are achieved.
- c. Advise the Engineer 48 hours in advance of each test. Failure to so notify will require test to be rescheduled.

2. Testing Requirements: Duct system shall be tested for air leakage in accordance with SMACNA "HVAC Air Duct Leakage Test Manual". See Section 15990 Testing, Adjusting, and Balancing for additional requirements.

3.4 ADJUSTING AND CLEANING

- A. Cleaning of General Service Ductwork: Remove debris from general-purpose ductwork and accessories, and then blow ductwork clean with air movement provided by the system fan or blower. Supply ductwork shall be blown clean before final branch connections are made to terminal units or before terminal grilles, registers, or diffusers are installed.
- B. Cleaning of Cleanroom Service Ductwork (SA Ducts, RA Ducts, Recovery Ducts): Remove dust and grease from cleanroom ductwork and accessories prior to delivery to the project site. Clean with a solution of DI water and isopropyl alcohol to remove dirt and grease. Wrap ductwork and components in plastic wrap prior to shipment from fabrication shop. Unwrap at the jobsite and install immediately. If installation is incomplete, wrap all open ends with plastic wrap to prevent dust entering ducts. Ductwork and components shall be maintained in a clean condition by the Contractor until air handling unit fans are operational.

3.5 SCHEDULE

- A. Fabricate portions of systems listed in Section 15840.01_dat from sheet metal with gauge and reinforcement in accordance with SMACNA duct construction tables.

END OF SECTION